

■ Hackathon Git + GitHub Team Workflow

3 people • 3 laptops • AI coding agents • 1 shared project

Goal: Teeno members parallel kaam karein, apni branches par code push karein, aur final working code ko safely **main** branch mein merge karein.

1. Tumhari understanding bilkul sahi hai

Haan. Basic workflow exactly ye hai: **repo clone** → **apni branch** → **AI se assigned work** → **add/commit** → **push** → **Pull Request** → **main mein merge**.

Ek correction: dost tumhara local **.git folder** clone nahi karte. Woh **GitHub repository** clone karte hain. Har laptop apni local copy rakhta hai aur GitHub common source of truth hota hai.

2. Final team structure

Member	Branch	Main responsibility
Kundan	backend	Backend / API / database
Friend 1	frontend	UI / pages / components
Friend 2	ai	AI / core logic / integrations

Visual:

GitHub main

■ backend → Kundan

↓ frontend → Friend 1

■ ai → Friend 2

Finished branches → Pull Request → **main**

3. STEP 0: Owner setup

Tumhare paas already local project + GitHub repo connected hai. Repository ko team ke members ke saath share karne ke liye unhe GitHub repository access/collaborator permission do.

GitHub → Repository → Settings → Collaborators / Access → Add people → friends ke GitHub usernames add karo.

Important: GitHub password ya tumhari private SSH key kisi friend ko mat do. Har member apne GitHub account ke saath apni SSH key authenticate kare.

4. STEP 1: Friends apne laptops par repo clone karein

Friend 1 aur Friend 2 ko tumhare laptop ka folder nahi chahiye. Unke laptops par terminal:

```
git clone git@github.com:Kundan-Maurya-0102/LpuHackathon.git
cd LpuHackathon
```

Ab unke paas project ki same starting files ki local copy hogi.

5. STEP 2: Har member apni branch banaye

Kundan:

```
git checkout -b backend
```

Friend 1:

```
git checkout -b frontend
```

Friend 2:

```
git checkout -b ai
```

Verify:

```
git branch  
# * tumhari-current-branch
```

6. STEP 3: AI agents ko controlled tasks do

AI IDE mein poora project ek saath banwane ke bajay har member ko clearly bounded responsibility do.

Frontend agent prompt idea: You own only frontend. Build the assigned UI. Do not modify backend or AI/core files unless explicitly requested.

Backend agent prompt idea: You own only backend/API/database. Do not modify frontend files.

AI agent prompt idea: You own AI/core logic and integrations. Avoid changing unrelated frontend/backend files.

Before coding, humans decide the folder structure and interfaces. AI agents should implement tasks inside those boundaries.

7. STEP 4: Work → Commit → Push

Example: Kundan backend par kaam complete karta hai:

```
git status
git add .
git commit -m "Build backend API"
git push -u origin backend
```

Friend 1:

```
git add .
git commit -m "Build frontend"
git push -u origin frontend
```

Friend 2:

```
git add .
git commit -m "Add AI integration"
git push -u origin ai
```

8. STEP 5: Merge into main

GitHub par Pull Request banao:

Pull requests → New pull request → base: main → compare: backend → Create pull request → review → Merge pull request → Confirm merge

Isi tarah frontend aur ai branches merge honghi.

Meaning: **backend** → **main** means backend branch ke changes main branch mein add karna.

9. ■■ Sabse important sync rule

Merge ke baad baaki members ko apni branch ko latest main ke saath update karna chahiye, especially jab shared files change hui hon.

```
git checkout main
git pull origin main
git checkout frontend
git merge main
```

Backend ke liye `frontend` ki jagah `backend`, AI ke liye `ai` use karo.

Agar branch mein uncommitted work hai, pehle commit/stash karke hi branch switch/merge karo.

10. Merge conflict kya hai?

Agar do members same file ki same lines ko different tareeke se edit karte hain, Git ko decide karna mushkil ho sakta hai. Isse merge conflict kehte hain.

Hackathon rule: responsibilities ko folders/files ke hisaab se separate rakho. Example: backend/ → Kundan, frontend/ → Friend 1, ai/ → Friend 2. Shared files jaise README, package config, routing entrypoints ko coordinate karke edit karo.

11. ■ Golden rules for hackathon

1. Main branch ko directly code mat karo.
2. Har member ki separate branch ho.
3. Ek file ko simultaneously 2 agents se edit mat karwao.
4. Har meaningful milestone ke baad commit + push.
5. AI agent ko clear scope do.
6. API keys / passwords / private SSH keys GitHub par kabhi commit mat karo.
7. Merge se pehle working feature test karo.
8. Main merge hone ke baad team ko sync karao.
9. Live demo ke saath backup video/screenshots rakho.
10. Hackathon rules check karo ki AI tools aur pre-built code ka kya policy hai.

12. Fast command cheat sheet

Clone:

```
git clone git@github.com:Kundan-Maurya-0102/LpuHackathon.git
```

New branch:

```
git checkout -b backend
```

Check branch:

```
git branch
```

Check changes:

```
git status
```

Save:

```
git add .  
git commit -m "message"
```

Push:

```
git push -u origin backend
```

Update main:

```
git checkout main  
git pull origin main
```

Update your branch from main:

```
git checkout backend  
git merge main
```

13. One complete example

- 1) Tum backend branch banate ho.
- 2) Friend 1 repo clone karke frontend branch banata hai.
- 3) Friend 2 repo clone karke ai branch banata hai.
- 4) Teeno AI agents ke saath parallel code karte hain.
- 5) Har member apne branch par commit + push karta hai.
- 6) Backend ready → backend → main PR → merge.
- 7) Frontend ready → frontend → main PR → merge.
- 8) AI ready → ai → main PR → merge.
- 9) Sab latest main pull/merge karke integrated project test karte hain.
- 10) Final demo + submission.

14. Emergency commands

Galat branch par ho?

```
git branch  
git checkout backend
```

Remote check:

```
git remote -v
```

Latest main lana:

```
git checkout main  
git pull origin main
```

Remote branches dekhna:

```
git branch -r
```

Recent commits:

```
git log --oneline --all --decorate -10
```

Final mental model: GitHub = common storage/source of truth. Branch = individual work lane. Commit = saved checkpoint. Push = apna checkpoint GitHub par bhejna. Pull Request = apna finished work main mein merge karne ki request. Main = final integrated project.

Official reference: [GitHub documentation on branches, pull requests, collaboration and SSH authentication.](#)